

		solving, $\mathcal{E} = 0.0 \text{ V}$
22	B	Since the galvanometer reads zero when PS = 40.0 cm p.d. across the wire PS = 5.00 mV. A length of 40.0 cm of wire PS has a resistance of $\frac{40.0}{100.0}(5.00) = 2.00 \Omega$. Using potential divider rule, $\frac{2.0}{5.0 + R}(2.000) = 5.00 \times 10^{-3}$ $R = 795 \Omega$
23	C	Since $\mathcal{E} = 1.5 \text{ V}$, $\mathcal{E} = 1.5 \text{ V}$